

Question 1 – Manufacturing Quality Inspection

Scenario

An automated inspection system is used in a manufacturing plant to examine metal parts for surface defects. During image acquisition, electrical disturbances from nearby equipment create repeated horizontal stripe patterns on the captured images, making defect identification difficult.

(a) Interpretation of the Scenario

In manufacturing industries, cameras are commonly used to inspect metal components for defects such as cracks, scratches, dents, and surface imperfections. However, electrical interference from machines or power sources may introduce repetitive horizontal stripe patterns into the captured images. This type of disturbance is called **periodic noise** because it repeats at regular intervals. If this noise is not removed, the inspection system may fail to detect defects correctly. Therefore, eliminating periodic noise is necessary to improve image quality and ensure reliable product inspection.

(b) Effect of Noise on Inspection Accuracy

Periodic noise has a negative impact on the performance of automated inspection systems.

- Fine cracks and scratches may become hidden.
- Repetitive stripe patterns interfere with edge detection.
- Image clarity and contrast are reduced.
- Defective products may be identified as normal products.
- False defect detection can increase production costs.
- Manual inspection may become necessary, reducing automation efficiency.

Hence, periodic noise lowers the overall accuracy and reliability of quality inspection.

(c) Recommended Enhancement Technique

The most suitable enhancement method is Fourier Transform with Notch Filtering.

Working Process:

1. Acquire the noisy image.
2. Transform the image into the frequency domain using the Fourier Transform.
3. Locate the frequency components responsible for periodic noise.
4. Apply a Notch Filter to suppress those unwanted frequencies.
5. Perform the Inverse Fourier Transform to reconstruct the cleaned image.

This technique removes repetitive stripe noise while preserving important image details required for defect detection.

(d) Justification

The Fourier Transform converts the image into its frequency components, making periodic noise appear as distinct peaks in the frequency spectrum. A Notch Filter selectively removes only those unwanted frequency components without affecting useful image information. As a result, important features such as cracks and scratches remain clear while the repetitive interference is eliminated. Therefore, this approach is highly effective for manufacturing quality inspection.

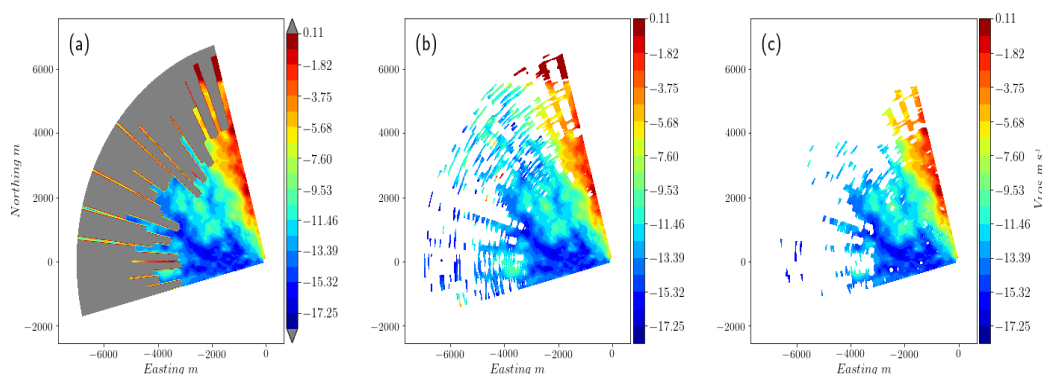
(e) Logical Sequence

1. Capture the image of the metal component.
2. Detect the presence of periodic stripe noise.

3. Apply the Fourier Transform.
4. Identify unwanted frequency components.
5. Remove them using a Notch Filter.
6. Perform the Inverse Fourier Transform.
7. Obtain an enhanced image for accurate defect detection.

Conclusion

Electrical interference can introduce periodic noise that significantly reduces inspection accuracy. Using the Fourier Transform together with Notch Filtering effectively removes repetitive noise while preserving important defect information. This improves image quality, increases detection accuracy, reduces false results, and enhances the efficiency of automated manufacturing inspection systems.



Question 2 – Autonomous Vehicle Navigation

Scenario

A self-driving vehicle captures road images while driving in heavy rain. The rainy conditions introduce unwanted noise into the images, but the system must still clearly identify lane markings and road boundaries to ensure safe and accurate navigation.

(a) Interpretation of the Scenario

Autonomous vehicles depend on cameras to recognize lane markings, road edges, traffic signs, pedestrians, and other vehicles. During heavy rainfall, raindrops and poor weather conditions introduce noise into the captured images, reducing their clarity. Even under these conditions, the vehicle must accurately detect road features to navigate safely. Therefore, an image enhancement technique is required that removes noise without affecting important edges and boundaries.

(b) Importance of Preserving Edges

Maintaining clear edges is essential for the proper functioning of autonomous driving systems.

- Lane markings can be detected more accurately.
- Road boundaries remain visible for safe navigation.
- Vehicles and pedestrians are identified more effectively.
- Traffic signs become easier to recognize.
- Sharp edges help prevent incorrect steering decisions.
- Overall driving safety and navigation performance are improved.

Therefore, preserving edge information is one of the most important requirements in autonomous vehicle vision systems.

(c) Recommended Filtering Technique

The most suitable filtering method is the **Bilateral Filter**.

Working Process:

1. Capture the noisy road image.
2. Analyze neighboring pixels based on their spatial distance.
3. Compare the intensity values of nearby pixels.
4. Smooth only those pixels that have similar intensity values.

5. Preserve edges while reducing unwanted noise.
6. Produce a cleaner image for further road analysis.

The Bilateral Filter effectively removes rain-induced noise while maintaining the sharpness of lane markings and road edges.

(d) Justification

The Bilateral Filter considers both the physical distance between pixels and the difference in their intensity values. Unlike traditional smoothing filters, it reduces noise without blurring important boundaries. As a result, lane markings, road edges, and traffic signs remain clear even after filtering. This makes the Bilateral Filter an ideal choice for autonomous vehicle image processing because it improves visibility while protecting critical image details.

(e) Logical Sequence

1. Capture the road image.
2. Detect the presence of rain-induced noise.
3. Apply the Bilateral Filter.
4. Reduce unwanted noise from the image.
5. Preserve lane markings and road edges.
6. Detect road boundaries.
7. Identify vehicles, pedestrians, and obstacles.
8. Perform lane detection for safe vehicle navigation.

Conclusion

Heavy rainfall introduces noise that can reduce the quality of road images captured by autonomous vehicles. Applying the Bilateral Filter effectively suppresses noise while preserving important edges and boundaries. This improves image clarity, enhances object and lane

detection, increases navigation accuracy, and ensures safer driving under adverse weather conditions.

